

What is virtualization?

Virtualization is a technology that allows us to run multiple virtual machines or operating systems on a single physical server. Instead of using separate physical servers for each operating system, we use a hypervisor to share the server's resources, such as CPU, RAM, storage, and network, among multiple virtual machines. This helps reduce hardware costs and makes better use of available resources. For example, VMware ESXi is a hypervisor used for virtualization.

Easy way to remember:

One physical server → Hypervisor → Multiple virtual machines → Better resource utilization.

How virtualization works?

Virtualization works by using a hypervisor, which is virtualization software installed on a physical server. The hypervisor sits between the physical hardware and the virtual machines. It manages and distributes resources such as CPU, RAM, storage, and network to multiple virtual machines, allowing several operating systems to run on one physical server. For example, VMware ESXi is a hypervisor.

Easy structure to remember:

Physical Server → Hypervisor → Virtual Machines → Operating Systems + Applications.

Key point: The hypervisor manages the hardware resources and allows multiple virtual machines to run at the same time.

Benefits of Virtualization

There are several benefits of virtualization. First, it reduces hardware and infrastructure costs because multiple virtual machines can run on one physical server. It also reduces data center space and power consumption. Virtualization makes management easier, provides better resource utilization and flexibility, and allows virtual machines to be deployed quickly using templates. It also supports redundancy and easier disaster recovery. In addition, snapshots can be useful for testing, patching, and making changes because you can return to an earlier VM state when appropriate.

Remember:

Cost → Space → Power → Management → Resources → Deployment → Redundancy → Testing.

Virtualization Basic Terms

1. What is a Bare Metal Server?

- A bare metal server is a physical server. It is the actual hardware, such as a Dell or HPE server, where an operating system or hypervisor can be installed.

2. What is a Hypervisor?

- A hypervisor is software that allows us to create and run multiple virtual machines on a physical server. It manages hardware resources such as CPU, memory, storage, and network. VMware ESXi is an example of a hypervisor.

3. What is a Host?

- A host is a physical server that runs the hypervisor and provides the resources used by virtual machines.

4. What is a Virtual Machine?

- A virtual machine, or VM, is a software-based computer that runs on a physical host. It can have its own operating system, applications, CPU, memory, storage, and network configuration.

5. What is vCenter?

- vCenter Server is a centralized management platform used to manage VMware hosts and virtual machines in a virtual infrastructure.

6. What is VDI?

- VDI stands for Virtual Desktop Infrastructure. It provides users with virtual desktops that run on servers instead of on their local physical computers.

7. What is P2V?

- P2V means Physical to Virtual. It is the process of converting or migrating a physical server into a virtual machine.

8. What is V2V?

- V2V means Virtual to Virtual. It is the process of migrating a virtual machine from one virtual environment or platform to another.

9. What is a VM Template?

- A VM template is a preconfigured image used to quickly create new virtual machines with a standard operating system and configuration.

10. What is a Snapshot?

- A snapshot captures the state of a virtual machine at a specific point in time. It can be useful before making changes so you can return the VM to an earlier state. However, a snapshot is not a replacement for a proper backup.

11. What is Cloning?

- Cloning is the process of creating a copy of an existing virtual machine. The new VM can then be used as a separate machine.

VMware / Virtualization Platforms

1. What are some common virtualization platforms?

“Some common virtualization platforms are VMware, Microsoft Hyper-V, Oracle virtualization, Citrix XenServer, and KVM.”

2. What is VMware ESXi?

“VMware ESXi is a hypervisor that runs on a physical server and allows multiple virtual machines to run on the same hardware.”

3. What is Microsoft Hyper-V?

“Hyper-V is Microsoft’s hypervisor and virtualization technology. It allows organizations to create and run virtual machines on physical servers.”

4. What is KVM?

“KVM stands for Kernel-based Virtual Machine. It is a virtualization technology built into the Linux kernel that allows Linux to act as a hypervisor.”

5. What is Oracle VirtualBox?

“Oracle VirtualBox is a virtualization application that can be installed on an existing operating system. It can be used to create and run virtual machines, especially for testing and learning.”

6. What is the difference between VMware ESXi and VirtualBox?

“VMware ESXi is a bare-metal hypervisor designed for servers and enterprise virtualization. VirtualBox is a hosted virtualization application that runs on top of an existing operating system.”

7. What is Azure?

“Azure is Microsoft’s cloud computing platform. It provides services such as virtual machines, storage, networking, and databases.”

8. What is AWS?

“AWS stands for Amazon Web Services. It is Amazon’s cloud computing platform that provides computing, storage, networking, and other cloud services.”

9. What is Google Cloud?

“Google Cloud is Google’s cloud computing platform. It provides services such as virtual machines, storage, networking, and databases.”

10. What is the difference between a hypervisor and a cloud platform?

“A hypervisor is software that creates and runs virtual machines on physical hardware. A cloud platform provides computing and other IT services over the internet. For example, ESXi is a hypervisor, while Azure and AWS are cloud platforms.”

Question: Which virtualization platform are you familiar with?

I am currently learning VMware virtualization, particularly ESXi and vCenter. I am studying how to create and manage virtual machines, allocate resources, and troubleshoot issues in a VMware environment.

Easy memory

VMware → ESXi → vCloud
Microsoft → Hyper-V → Azure
Oracle → VirtualBox / OLVM → OCI
Linux → KVM → Redhat Cloud Suite
Amazon Cloud → AWS
Google Cloud → Google Cloud